1 FF++ triplet, drawn on the fly



genuine beautified forged

A scripted beautification and a self-blend or observed forgery from one genuine frame; **the forged item is beautified too**, half of the time.

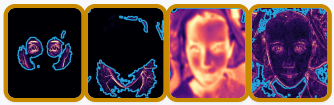
2 Commercial render split into single operations



original O render R

warp f: optical flow $O \to R$

residual $P = R - O(x+f)$



eye contour tone texture

Warps of O by f (eyes, elsewhere) and the low- and high-pass parts of P, shown as $|C - O|$ with the exact footprint M outlined; in sequence they rebuild R at 108 dB. **18,734 renders** (Tencent, Megvii) give **50,112** items; Alibaba is held out for testing.

3 Part-level forgery, exact M



original x_0 donor nose footprint M

61,510 on FF++ frames: eyes, nose or mouth from a donor or inpainted; outside M every pixel is the original's.

4 One backbone, its heads

CLIP ViT-L/14
frozen, LoRA rank 16 on all attention projections: 3.2 M of 306 M trained

browser: EfficientNet-B4, 17.6 M, same heads

Class logits f (forged), b (beautified): $P(\text{forged}) = \sigma(f)$, $P(\text{beautified}) = (1-\sigma(f))\,\sigma(b)$, $P(\text{real}) = (1-\sigma(f))(1-\sigma(b))$. The forgery score never reads b.

Operation head: eye · contour · tone · texture present or not (BCE on beautified items).

Evidence head: 1 × 1 conv on the 16 × 16 tokens, trained against M (BCE + Dice). The classifier does not read it, so (5) tests it.

Loss = BCE(f) + BCE(b) + BCE(ops) + 2 L1(amounts) + BCE(m, M) + Dice(m, M); the auxiliary amount head and the encoder are drawn in Figure 5. AdamW 10^{-4}, 8 epochs; checkpoint chosen on validation data only.

Main model, held-out



forged 1.00
evidence: **nose**
operations all ≤ 0.23

FF++ test frame, nose from a donor



beautified 0.98
evidence: **eyes**
eye 0.97, others ≤ 0.01

Alibaba eye render, a service never used in training

5 Three tests that could fail

Exposure which class saw smoothing	Smoothing seen by forged faces only: **54%** of smoothed genuine faces accused; by both classes: **12%**; by genuine only: deepfakes escape.
Restoration is the mark the reason?	Original pixels put back inside the marked region remove **94%** of part-level forgery decisions; a post-hoc heat map: about half.
Matching does text carry the image?	A judge pairs a VLM description with its own photo about as often as a **coin flip**; so code writes the sentences and a rule checker matches each to the record.

6 Browser

MediaPipe landmarks → crop → **ONNX Runtime Web** (EfficientNet-B4, 67 MB) → contour, part, checked sentences.

0.59 s per photo on an iPhone, 0.51 s on a desktop; nothing uploaded.

training and model · outputs · tests · code on the device · exact footprint M